PPOC pediatric EHR snapshot: an exploratory data analysis

A project-neutral reference for anyone analysing this extract. Every figure is measured from the delivered bundle; the report states what the data support, what they do not, and which checks this extract cannot answer at all.

0. How to use this report

Three ways in, depending on what you came for.

0.1 Three ways in

This report describes one snapshot of one pediatric primary-care EHR extract. It belongs to no project: it states what the data are, what they support, and what they cannot answer, and it leaves the research question to you.

Where to start

IF YOU ARE	START HERE
New to this extract	Read Part 1, then the not-applicable table in Part 2. Twenty minutes, and it will save days.
About to use a specific field	Find it in the Part 6 field index, then follow the finding it links to.
Explaining a number that looks wrong	Check the Part 7 artifact catalogue before assuming a bug in your code.
Planning a study	Part 1.4 first. The cohort selection invalidates several whole classes of question, and it is not visible in any field.

Every number here was measured from the delivered bundle for snapshot `2026-08-24`; none is copied from another document without being recomputed. The cohort is pinned to 31 Dec 2024 and the extract was cut on 03 Feb 2025.

> **What this report is not.** It is not a clinical validation, not a registered analysis, and not a statement about any individual child. Every figure is an aggregate, and any cell resting on fewer than 10 records is suppressed.

1. The snapshot

What this extract contains, how it was built, and what its construction forecloses.

1.1 Package identity and integrity

Everything in this report was computed from the typed DuckDB bundle of package `ppoc-pediatric-ehr` 1.0.0, snapshot `2026-08-24`, sha256 `425c6f873cefc149344570561a03b33c69a6a6af7fa18bc777c0429579507116`. The bundle is opened read-only and is never copied into this repository.

Three independent sources state how large this extract should be: the bundle manifest, the PPOC delivery documents committed under `docs/`, and the data itself. They are reconciled here before any other figure is computed, so that a bundle drawn from a different extract would be visible rather than silently profiled.

Row counts, measured against both declared sources

RESOURCE	MEASURED	BUNDLE MANIFEST	PPOC DOCUMENT	AGREES
patients	250,588	250,588	250,588	yes
patients_augmented	250,588	250,588	—	yes
visits	6,494,473	6,494,473	6,494,473	yes
visits_augmented	6,494,473	6,494,473	—	yes
labs	17,230,681	17,230,681	17,230,681	yes
medications	3,823,049	3,823,049	3,823,049	yes
problem_list	1,709,584	1,709,584	1,709,584	yes
referrals	349,827	349,827	349,827	yes

Distinct patients per resource, against the PPOC counts

RESOURCE	MEASURED	PPOC DOCUMENT	AGREES
visits	250,588	250,588	yes
problem_list	238,823	238,823	yes
labs	247,271	247,271	yes
medications	236,323	236,323	yes
referrals	138,071	138,071	yes

The lab resource carries a second stated figure: 6,578,838 distinct lab orders behind the resulted components. The bundle holds 6,578,838. Across every count above, 0 disagree.

1.2 Resource map, grain, and keys

The extract is 8 tables carrying 254 columns between them. Grain matters more than row count here: three of the resources are keyed on something other than the patient or the visit, and one of them needs two columns to be unique.

The eight resources

RESOURCE	ROWS	COLS	GRAIN	PRIMARY KEY	LINKS TO
patients	250,588	11	one row per patient	patient_id	—
patients_augmented	250,588	87	one row per patient	patient_id	patients
visits	6,494,473	43	one row per patient per encounter	visit_id	patients
visits_augmented	6,494,473	82	one row per patient per encounter	visit_id	visits
labs	17,230,681	12	one row per resulted component of a lab order	lab_order_id + result_line_num	patients; visits (partial)
medications	3,823,049	8	one row per medication order or historical record	med_record_id	patients; visits (partial)
problem_list	1,709,584	5	one row per problem-list entry	problem_list_id	patients
referrals	349,827	6	one row per referral order	referral_id	patients; visits (partial)

`visit_id` on labs, medications, and referrals is a partial link by design, not a defect: an order placed outside a visit carries an identifier that resolves to no encounter in this extract. Section 3.2 measures how partial.

1.3 The two layers, and where they disagree

Every visit and every patient appears twice: once in the raw extract as PPOC delivered it, and once in an augmented layer that adds CDC-derived z-scores, percentiles, velocities, and flags. Where a field exists in both, the two should agree. Across all 6,494,473 joined visit rows, five of the six shared fields do.

Shared visit fields, raw against augmented

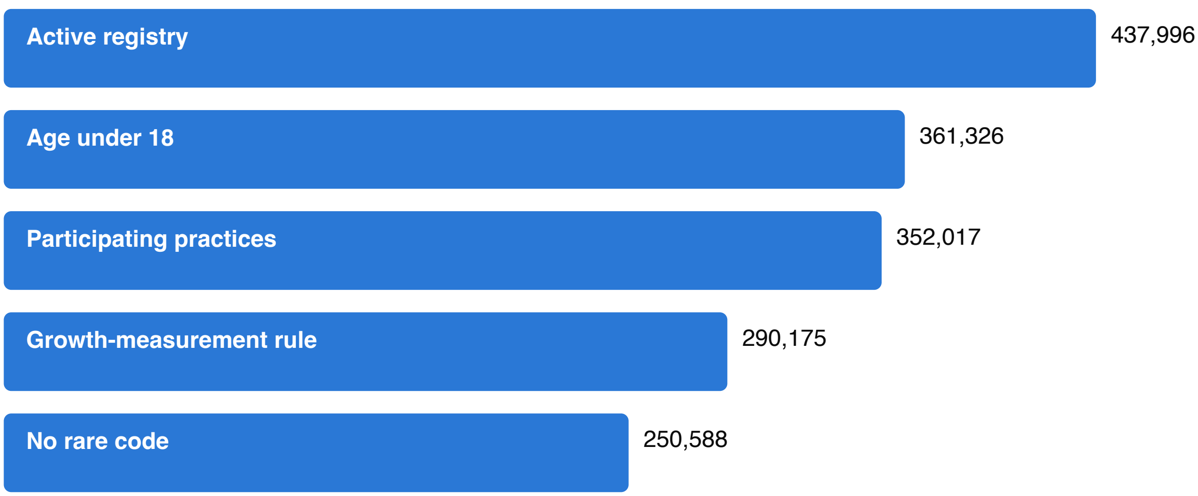
FIELD	ROWS DIFFERING	SHARE
height_in	0	0.00%
weight_oz	0	0.00%
head_circ_cm	0	0.00%
encounter_type	0	0.00%
age_in_days	0	0.00%
BMI / bmi	3,658,277	56.33%

BMI is the exception, and the disagreement is structured rather than noisy. 1,703,005 visits carry a raw `BMI` where the augmented `bmi` is null, at a median age of 0.51 years; the augmented layer withholds BMI below age 2, where a CDC BMI-for-age reference does not apply, while the raw value is computed inside the source EHR at every age. A further 41 rows go the other way, and 536 carry both values differing by more than 0.01.

> **Implications for analysis.** Reading `visits.BMI` silently yields infant BMI values that the augmented layer deliberately suppresses, and the two layers will not reproduce each other's descriptive statistics. Choose a layer for a stated reason and record which; do not mix them within one analysis. The 536 rows where both are present and disagree are small enough to screen individually.

1.4 How this cohort was built

This is the most consequential section of the report, because it describes a property of the data that no field exposes and that no amount of analysis can recover. The 250,588 patients here are what remains after four successive exclusions applied by PPOC to an active-patient registry of 437,996. The final count reconciles against the delivered data exactly: 250,588 patient rows.



Cohort construction, registry to delivered extract

The four exclusions

STEP	CRITERION	EXCLUDED	REMAINING
0	On the PPOC active-patient registry		437,996
1	Age under 18 as of 31 Dec 2024	76,670 excluded	361,326
2	Excluding 2 practices that declined participation	9,309 excluded	352,017
3	At least 5 growth measurements of one type on distinct dates, spanning over 1095 days, last measurement within 400 days	61,842 excluded	290,175
4	Carrying no rare diagnosis, medication, or lab	39,587 excluded	250,588

"Active" on that registry means living status alive, not flagged as a test or inactive record, an active PPOC primary-care association, and either a visit in the last three years or one scheduled in the next fifteen months. The cohort is pinned to 31 Dec 2024 and the extract was cut on 03 Feb 2025.

The fourth exclusion is the one most likely to be missed, because it removed *patients* rather than codes. A diagnosis, medication, or lab occurring fewer than 11 times in the data set was classed rare, and every patient carrying one was dropped.

What the rarity exclusion removed

VOCABULARY	DISTINCT VALUES	CLASSED RARE	SHARE
ICD-10 diagnosis codes	30,493	18,604	61%
Simple generic medications	2,503	1,391	56%
Lab procedures	13,402	9,621	72%

> **Implications for analysis.** Rare conditions, rare exposures and uncommon labs are absent by construction, not merely sparse: a study of any of them returns a confident low rate rather than an obviously missing population. 61% of diagnosis codes, 56% of medications and 72% of lab procedures left with their patients. Because the registry requires living status alive, there are no deceased patients and mortality is not an available outcome. Because entry required at least five growth measurements, trajectory richness is an entry criterion and not a finding about pediatric care. And because the last measurement had to fall within 400 days of the cohort date, the panel is right-censored by design. No frequency in this extract is a population prevalence.

Two ambiguities in the source documents are recorded rather than silently resolved. The cohort workbook describes the under-three exemption as applying to the span requirement for children who already have five measurements, while the extract diagram describes it as age under three with at least one measurement. The same two documents give the rarity threshold as "fewer than 11 occurrences" and "under 10 patients".

1.5 The de-identification envelope

`age_in_days` is the only clock. The extract carries no calendar date, no time of day, no site, practice, provider, or geography, and no free text from any note. That is stated once here and referenced from Part 2 rather than re-argued at each check it rules out.

Checks this extract forecloses, and why

STANDARD CHECK	WHY IT CANNOT BE RUN
Duplicate-patient detection	no name, birth date, or linkage key survives
Batch-entry clustering	ages are integer days; there is no time of day
System downtime gaps	no calendar axis on which a void could appear
Missingness by site or provider	no such column exists in any resource
Calendar trend breaks and policy shifts	no calendar axis
Copy-forward of note text	no note text is included
Documentation timing	no timestamps

One qualification, because "no calendar axis" is easy to overstate: the cohort itself is pinned to 31 Dec 2024 and the extract was cut shortly after. Ages are relative to each child's birth, but the *window* is fixed and known, which is what makes the recency criterion in 1.4 a right-censoring rule rather than an unknown.

2. Checklist coverage

Every item of the general EHR EDA checklist, mapped to what this snapshot can and cannot support.

2.1 The checklist, item by item

This part exists so that nobody has to wonder whether a standard check was skipped or was impossible. Of 44 items in the general EHR exploratory-analysis checklist, 31 are covered here, 4 are partially covered, and 9 cannot be run against this extract at all.

Checklist coverage

CHECKLIST SECTION	ITEM	STATUS	WHERE, OR WHY NOT
0 Provenance	Extraction window	covered	Cohort and extract dates recovered from the delivery documents — 1.4
0 Provenance	Inclusion/exclusion logic	covered	The full four-step funnel — 1.4
0 Provenance	Vendor, version, migration events	covered	Epic against converted legacy records — 3.7
0 Provenance	Data dictionary present	covered	Committed under docs/, reconciled field by field — 1.1
0 Provenance	Raw vs CDM vs custom extract	covered	A custom extract plus a derived augmentation layer — 1.3
1 Structural	Row and table counts	covered	Against the manifest and the vendor's own counts — 1.1
1 Structural	Primary key uniqueness	covered	All eight resources — 3.1
1 Structural	Referential integrity	covered	3.2
1 Structural	Duplicate patient detection	not applicable	No name, birth date, or linkage key survives de-identification — 1.5
1 Structural	Schema drift	covered	Live schema against the dictionary; three documented fields absent — 1.1
1 Structural	Grain per table	covered	Including that patient and age is not unique in visits — 3.1
2 Temporal	Timestamp semantics	covered	3.3
2 Temporal	Impossible sequences	covered	3.3
2 Temporal	Batch-entry clustering	not applicable	Ages are integer days; there is no time of day — 1.5
2 Temporal	System downtime gaps	not applicable	No calendar axis — 1.5
2 Temporal	Coding or vendor transition	partial	Epic against converted is computable; ICD-9 to ICD-10 is not, without dates
2 Temporal	Age sanity	covered	3.3
3 Missingness	Missingness per field	covered	3.4 and the field index

CHECKLIST SECTION	ITEM	STATUS	WHERE, OR WHY NOT
3 Missingness	Missingness pattern	covered	By age, sex, and encounter — 3.4
3 Missingness	Sentinel values	covered	3.5
3 Missingness	Not measured vs measured negative	covered	Two fields whose nulls carry meaning — 3.5
3 Missingness	Missingness by site or provider	not applicable	No site, department, or provider column exists — 1.5
4 Distributional	Univariate distributions	covered	4.3
4 Distributional	Unit inconsistencies	covered	4.3 and 4.4
4 Distributional	Digit preference and rounding	covered	4.2
4 Distributional	Categorical value counts	covered	3.6
4 Distributional	Outlier detection	covered	Bounds reported before any exclusion is recommended — 4.3
4 Distributional	Cross-field plausibility	covered	Raw against augmented layers — 1.3
5 Terminology	Code system vintage	covered	3.6
5 Terminology	Granularity consistency	covered	3.6
5 Terminology	Problem list staleness	covered	3.5
5 Terminology	Free text vs structured	covered	Laboratory result values are semi-structured text — 3.6
5 Terminology	Local or custom codes	covered	3.6
6 Workflow	Copy-forward detection	partial	Detectable on measurements; no note text is included
6 Workflow	Template or boilerplate detection	not applicable	No note text — 1.5
6 Workflow	Documentation timing	not applicable	No timestamps — 1.5
6 Workflow	Order/result reconciliation	covered	3.6
7 Population	Cohort representativeness	partial	The cohort is not representative and 1.4 says exactly how; no external benchmark ships with this repository
7 Population	Encounter type mix	covered	3.7
7 Population	Follow-up time distribution	covered	4.1
7 Population	Site or provider volume	not applicable	No such field — 1.5
8 Longitudinal	Calendar trend breaks	not applicable	No calendar axis. Age-axis profiles are reported instead and are not the same thing — 1.5
8 Longitudinal	Guideline or policy shift	not applicable	Requires calendar time — 1.5
8 Longitudinal	Vendor changeover effects	partial	The Epic against converted contrast only

The not-applicable list is the part worth reading before you start. Every entry is a consequence of de-identification or of what the extract simply does not carry, and no amount of analysis recovers any of them.

Checks this extract cannot support

CHECK	WHY
Duplicate patient detection	No name, birth date, or linkage key survives de-identification — 1.5
Batch-entry clustering	Ages are integer days; there is no time of day — 1.5
System downtime gaps	No calendar axis — 1.5
Missingness by site or provider	No site, department, or provider column exists — 1.5
Template or boilerplate detection	No note text — 1.5
Documentation timing	No timestamps — 1.5
Site or provider volume	No such field — 1.5
Calendar trend breaks	No calendar axis. Age-axis profiles are reported instead and are not the same thing — 1.5
Guideline or policy shift	Requires calendar time — 1.5

Implications for analysis. Treat the second table as a design constraint rather than a gap to work around. A protocol that depends on provider variation, time-of-day effects, calendar trends, or deceased patients cannot be run on this extract, and discovering that after cohort construction is expensive.

3. Integrity

Keys, linkage, the age axis, missingness, terminology, and capture.

3.1 Keys, grain, and uniqueness

Every declared primary key holds. The labs resource needs all three of its declared columns to be unique, which is worth stating because joining on order and component alone will multiply rows.

Declared keys, measured

RESOURCE	KEY	ROWS	DISTINCT KEYS	UNIQUE
patients	patient_id	250,588	250,588	yes
patients_augmented	patient_id	250,588	250,588	yes
visits	visit_id	6,494,473	6,494,473	yes
visits_augmented	visit_id	6,494,473	6,494,473	yes
medications	med_record_id	3,823,049	3,823,049	yes
problem_list	problem_list_id	1,709,584	1,709,584	yes
referrals	referral_id	349,827	349,827	yes
labs	lab_order_id + component + line	17,230,681	17,230,681	yes

What is *not* a key is the combination a longitudinal analysis reaches for first. 5,478 patient-days (0.08% of 6,488,911) carry more than one visit, covering 11,040 visit rows (0.17% of all visits). `age_in_days` is therefore not unique within a patient.

> **Implications for analysis.** Any trajectory ordered by age alone has ties, and any window function partitioned by patient and ordered by age will resolve them arbitrarily unless you say how. Decide whether to take the first row, the mean, or the non-null value, and apply it before the analysis rather than inside it.

3.2 Referential integrity and cross-resource linkage

`patient_id` resolves everywhere. `visit_id` does not, and the shortfall is large enough that treating it as a complete foreign key will quietly drop or duplicate rows.

Visit linkage by resource

RESOURCE	ROWS	VISIT_ID NULL	POPULATED BUT UNRESOLVED	SHARE OF POPULATED	UNRESOLVED PATIENT_ID
labs	17,230,681	0.00%	5,201,657	30.19%	0

RESOURCE	ROWS	VISIT_ID NULL	POPULATED BUT UNRESOLVED	SHARE OF POPULATED	UNRESOLVED PATIENT_ID
medications	3,823,049	0.00%	1,592,437	41.65%	0
referrals	349,827	7.10%	98,623	30.35%	0

This is documented behaviour rather than corruption. The data dictionary states for each of these resources that the visit link "may not match to all" when the order was placed or the record documented outside a visit. The trap is that the column is populated on nearly every row, so a required-looking key silently fails to join.

> **Implications for analysis.** Join to visits with an explicit outer join and count what fails, rather than an inner join that hides the loss. Anything computed per visit — encounter type, visit-level anthropometrics — is unavailable for the unresolved share, and that share is not random: it concentrates in orders placed outside encounters.

3.3 Age-axis consistency and impossible sequences

Age in days is the only clock, so ordering violations within a resource are visible directly. Counts below 10 are suppressed.

Ordering and range checks

CHECK	VIOLATING ROWS	ROWS CHECKED	SHARE
Lab result age earlier than lab order age	583,055	14,947,495	3.901%
Medication start age earlier than order age	329,107	3,539,983	9.297%
Medication end age earlier than start age	12,709	3,179,759	0.400%
Problem resolved age earlier than noted age	0	754,996	0.000%
Problem noted before birth	650	1,702,300	0.038%
Lab ordered before birth	47	17,230,681	0.000%
Medication ordered before birth	—	3,823,049	—
Visit recorded before birth	0	6,494,473	0.000%

An em dash in the violating-rows column means the count is nonzero but below the suppression threshold.

The lab and medication violations are the substantial ones, and both are documented at source. For a historically documented medication the order date is the date the record was *written*, not when the drug was started, and a charted approximation such as a month with no day is stored as the first of that month. End dates may sit in the future while a medication is active. Lab result and order ages derive from different source timestamps.

> **Implications for analysis.** Differences between two age fields in these resources are not reliable durations. Where you need an interval, take it from a single field across rows rather than between two fields on one row, and exclude historically documented medication records from any start-to-end calculation.

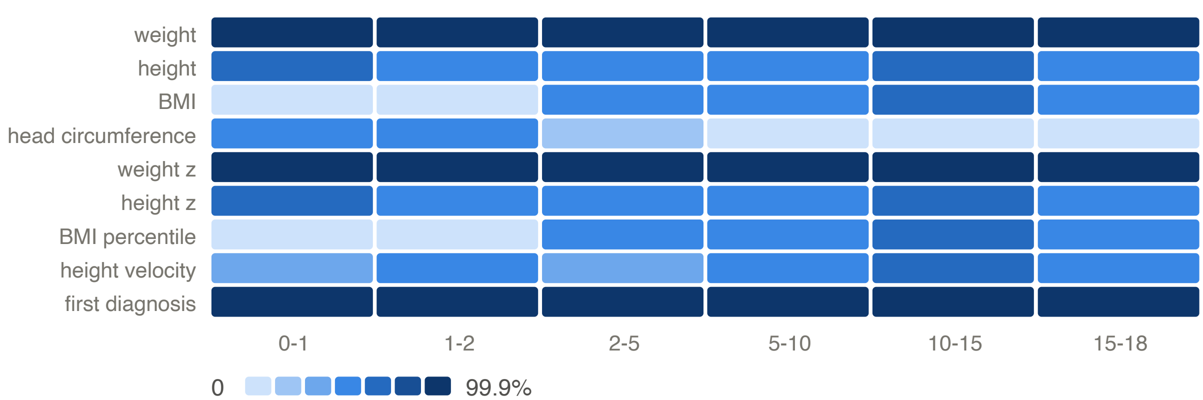
3.4 Missingness, by field and by age

Population was measured for all 176 columns in the extract, counting the repeated diagnosis and race families once each. 0 columns are entirely empty. The full table is Part 6; the sixteen least-populated columns are below.

The least-populated columns

RESOURCE	FIELD	POPULATED ROWS	MISSING
patients_augmented	dx_age_years_e24	1	100.0%
patients_augmented	dx_age_years_e72_11	1	100.0%
patients_augmented	dx_age_years_n25_0	1	100.0%
patients_augmented	dx_age_years_q78_1	2	100.0%
patients_augmented	dx_age_years_e22_0	3	100.0%
patients_augmented	dx_age_years_q78_0	10	100.0%
patients_augmented	dx_age_years_q77	15	100.0%
patients_augmented	dx_age_years_q87_4	17	100.0%
patients_augmented	dx_age_years_q98_5	17	100.0%
patients_augmented	dx_age_years_q98_0	26	100.0%
patients_augmented	dx_age_years_e23_6	31	100.0%
patients_augmented	dx_age_years_q87_2	32	100.0%
patients_augmented	dx_age_years_q96	36	100.0%
patients_augmented	dx_age_years_q98_4	42	100.0%
patients_augmented	dx_age_years_q87_3	46	100.0%
patients_augmented	dx_age_years_p04_3	53	100.0%

A single missingness rate hides the thing that matters most for a longitudinal extract: whether a field is missing *at random* or missing *by age*. For the measurement channels it is emphatically the latter.



Share of visits carrying each measurement, by age band

> **Implications for analysis.** Head circumference is an infant measurement and effectively disappears after age 2; BMI and its percentile are withheld below age 2 by the augmentation; height is recorded far less often than weight at every age. Any cohort defined by "has a complete measurement row" is therefore an age-selected cohort, and any model that drops incomplete rows inherits that selection. Report availability by age band before interpreting any age-stratified contrast.

3.5 Nulls that are not missing, and sentinels that are not data

Two of the largest null populations in this extract are not missing data at all, and reading them as missing throws away the majority of the signal in their columns.

Laboratory result flags

RESULT_FLAG	ROWS	MEANING
null	15,550,985	normal result
Abnormal	704,327	abnormal
High	513,650	abnormal
Low	361,300	abnormal
Sensitive	62,794	abnormal
Resistant	10,278	abnormal
High Panic	9,744	abnormal
(NONE)	5,881	abnormal

The data dictionary defines `result_flag` as an HL7 abnormality category in which the value `(NONE)` means a normal result and anything else means abnormal. This extract contains 5,881 literal `(NONE)` values and 15,550,985 nulls — 90.3% of all lab rows. The sentinel became a null somewhere between the source system and delivery, so **a null flag means normal, not unknown**.

`problem_list.resolved_date_age_in_days` behaves the same way: the dictionary defines null as "problem currently active". 951,677 of 1,709,584 entries (55.7%) are null, which is a statement about 56% of problems being open, not about missing dates.

Zero and blank values checked as possible sentinels

RESOURCE	FIELD	PATTERN	ROWS
visits_augmented	height_in	zero height	0
visits_augmented	weight_oz	zero weight	0
visits_augmented	head_circ_cm	zero head circumference	—
labs	result_value	empty result string	0
patients	sex	blank sex	0
patients	race_1	blank race_1	8,818
patients	ethnicity	blank ethnicity	5,464

> **Implications for analysis.** Never impute or drop on `result_flag` or `resolved_date_age_in_days` nullity. An abnormal-result rate computed as "non-null flags over non-null flags" will read as 100%; the correct denominator is all resulted rows. A problem-list resolution rate must count nulls as unresolved rather than excluding them.

3.6 Code systems, free text, and categorical hygiene

Diagnosis coding is almost entirely well-formed ICD-10. Of 14,714,503 filled encounter-diagnosis slots across 8,029 distinct codes, 145,992 (0.99%) do not match the ICD-10 shape; of 1,709,584 problem-list entries across 4,739 codes, 39,860 (2.33%).

The non-conforming diagnosis values

VALUE	SLOTS
IMO0002	116,950
U07.1	25,483
IMO0001	3,443
U09.9	77
U07.0	39

These are proprietary placeholders the source EHR emits when a clinical term has no ICD-10 equivalent. They carry no diagnostic meaning and should be excluded from code-based cohort definitions rather than treated as unmapped diagnoses.

Laboratory results are the opposite case. `result_value` is a text column: of 17,230,681 rows, 7,621,449 (44.2%) parse as a number and 2,494,261 (14.5%) are empty. Among the rest, 487,168 are censored results carrying a comparator prefix, and the remainder are qualitative results, specimen descriptors, and administrative non-results. A LOINC code is present on only 7.8% of rows.

The declared key holds, but 33,879 order-and-component pairs appear on more than one result line and 23,679 of those (69.9%) carry disagreeing values. The data dictionary records the cause: a result may fail to link back to its original order, which duplicates the record.

Categorical vocabularies before and after normalising case and internal whitespace

RESOURCE	FIELD	DISTINCT VALUES	AFTER NORMALISING	COLLAPSED
labs	lab_procedure_name	3,742	3,739	3
medications	med_simple_generic_name	1,073	1,073	0
referrals	requested_specialty	119	119	0

> **Implications for analysis.** A naive numeric cast on `result_value` silently discards more than half the populated values and turns a left-censored result into a missing one rather than a bound. Join labs on order, component *and* line number, or the duplicate lines will multiply rows and pick a value arbitrarily. The categorical vocabularies barely collapse under normalisation, so grouping by them is safe after trimming.

3.7 Capture: measurement presence is not measurement occurrence

Completeness by age says how often a column is filled. Encounter type says whether filling it could have meant a measurement.

Measurement and diagnosis presence by encounter type

ENCOUNTER TYPE	VISITS	WEIGHT PRESENT	HEIGHT PRESENT	FIRST DIAGNOSIS
Office Visit	4,725,643	99.9%	52.4%	95.8%
Well Visit (Conv.)	778,452	99.9%	98.7%	94.4%
Sick	580,991	99.9%	16.9%	95.4%
Follow-Up	92,370	99.8%	24.9%	93.1%
Walk-In	79,679	99.9%	8.7%	96.9%
Consult	32,355	99.9%	53.7%	99.6%
Conversion Encounter	32,007	99.9%	66.9%	17.3%
Newborn	31,142	99.9%	87.5%	99.1%
Telemedicine	25,658	97.4%	44.8%	99.8%
Telephone	22,053	99.3%	50.1%	34.8%
Weight Check	16,295	99.9%	37.7%	96.4%
Clinical Support	15,347	99.3%	15.4%	83.4%
Documentation	13,107	98.3%	84.4%	8.6%
Immunization	11,774	97.6%	30.7%	93.4%

Encounter types with fewer than 10,000 visits are omitted.

Telephone encounters carry a weight on 99.3% of 22,053 visits. A weight cannot be measured over the telephone, so those values were produced some other way — reported by a caregiver, carried from a nearby in-person encounter, or attached to an encounter whose type label does not describe how the patient was seen. Which of those it is cannot be determined from this extract.

Recording completeness by source system

ENCOUNTER SOURCE	VISITS	HEIGHT PRESENT	FIRST DIAGNOSIS
Epic	4,149,865	51.4%	99.7%
converted from a legacy system	2,344,608	58.7%	86.1%

The source-system split is the migration signal. Records converted from the practice network's previous EHR carry a first diagnosis on only 86.1% of encounters, which the data dictionary anticipates: converted encounters may be missing diagnosis information depending on the quality of the conversion.

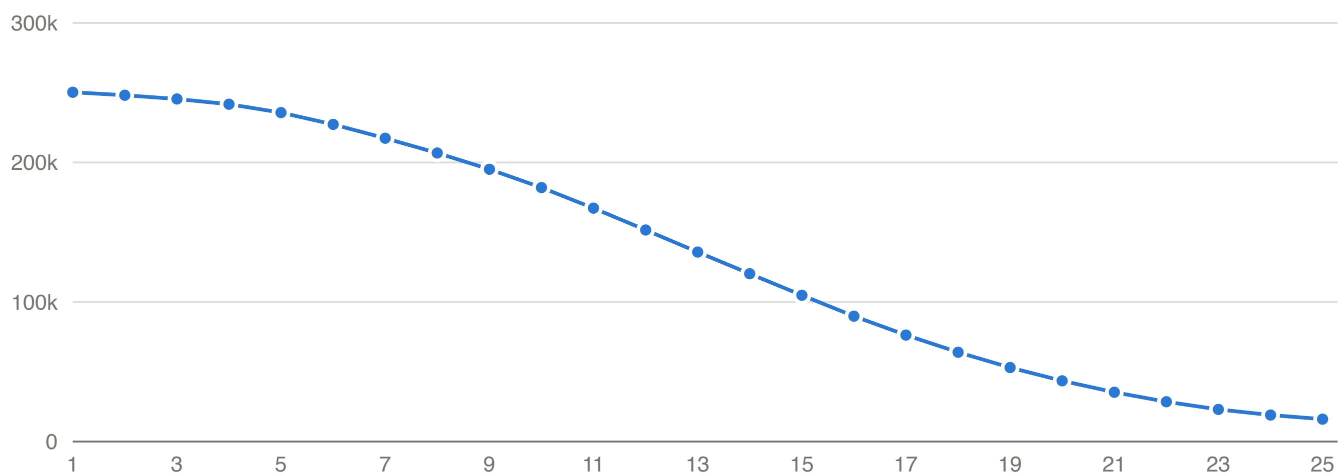
> **Implications for analysis.** A visit-level indicator that a measurement is present is not evidence that a measurement was taken at that encounter. If your design counts measurement occasions — visit density, monitoring intensity, follow-up adherence — restrict to encounter types where physical measurement is possible rather than relying on presence. And any diagnosis-based rate computed across the whole extract mixes two populations with very different coding completeness.

4. Anthropometrics

The richest and most artifact-prone measurements in the extract.

4.1 Trajectory supply: how many heights each child has

250,267 of 250,588 patients (99.9%) carry at least one derived height, 235,651 (94.0%) carry five or more, and 182,037 (72.6%) carry ten or more.



Patients retaining at least k height observations

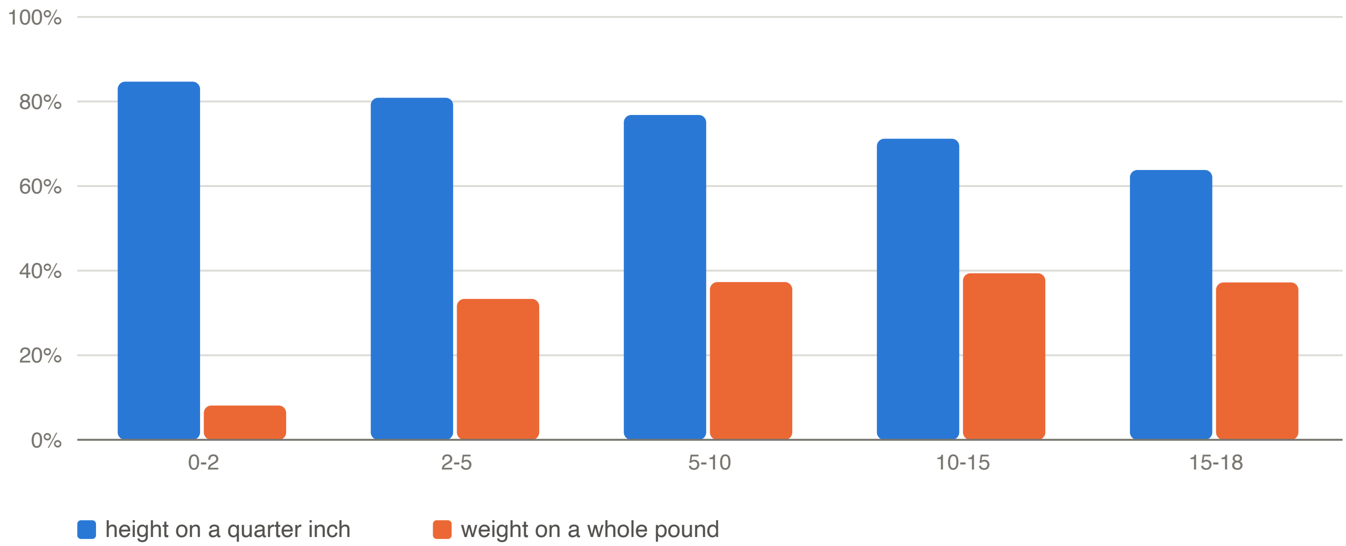
Height observations per patient

	AT LEAST K HEIGHTS	PATIENTS	SHARE OF COHORT
	1	250,267	99.9%
	3	245,449	97.9%
	5	235,651	94.0%
	10	182,037	72.6%
	15	104,838	41.8%
	20	43,501	17.4%
	25	16,040	6.4%

Implications for analysis. Read this against 1.4 before treating it as a fact about pediatric care. Cohort entry required at least five growth measurements of *some* type, so a dense height series here is partly the selection rule and partly the underlying practice; the two cannot be separated within this extract. What the curve does support is a feasibility estimate: how many children remain if your design needs k observations.

4.2 Recording units and the measurement grid

Height and weight are captured in imperial units and the metric columns are exact conversions of them. The recorded values are heaped on human-readable fractions: of 3,509,633 heights, 31.0% fall on a whole inch, 54.9% on a half inch and 80.0% on a quarter inch. Of 6,488,028 weights, 54.4% fall on a whole ounce and 24.6% on a whole pound.



Share of measurements falling on the coarse grid, by age

The two channels age in opposite directions. Height stays on its quarter-inch grid throughout childhood, while weight moves from ounce-level precision in infancy to whole pounds in adolescence, so the effective resolution of the weight channel degrades as children get older.

> **Implications for analysis.** One quarter inch is 0.635 cm, and the derived `height_cm` carries two decimals it has not earned. Any change smaller than roughly half the rounding interval is not distinguishable from the rounding itself, which sets a floor on the smallest trajectory deflection that can be detected at all. State the assumed precision wherever a measurement is written out, and set detection thresholds at or above the grid.

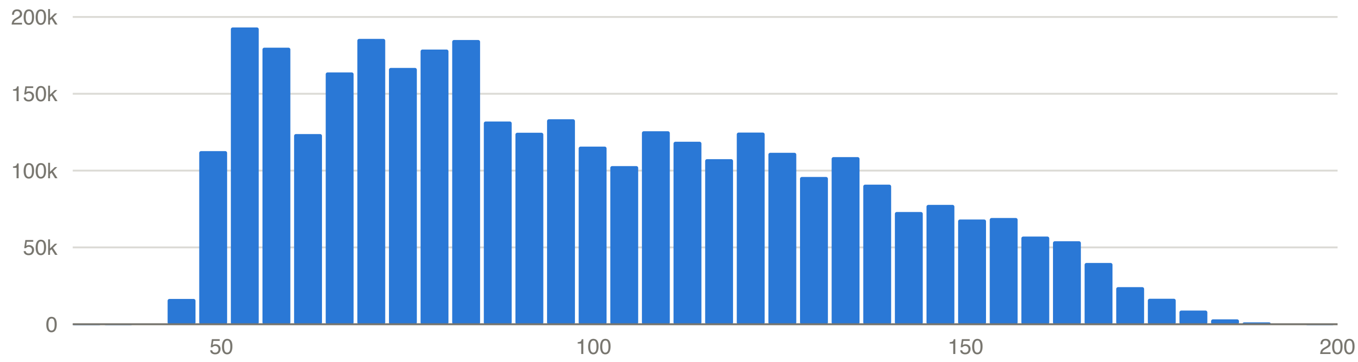
4.3 Distributions and plausibility bounds

The four measurement channels, summarised on the derived metric columns. The final column counts values outside a conventional review range; those are reported, not removed, because the decision to exclude belongs to the analysis rather than to this report.

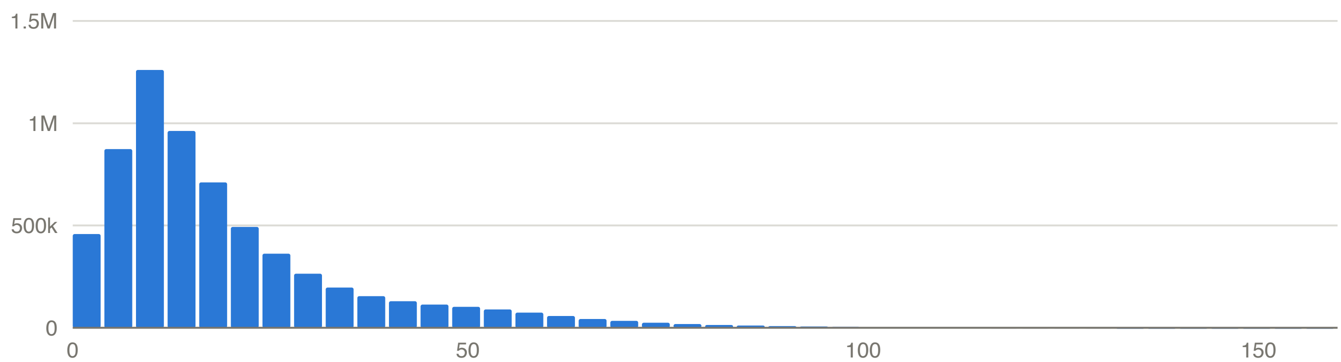
Measurement channels

CHANNEL	UNIT	VALUES	MIN	1ST PCT	MEDIAN	99TH PCT	MAX	OUTSIDE REVIEW RANGE
height	cm	3,491,662	37.47	48.26	92.71	173.41	196.60	0
weight	kg	6,483,007	0.01	2.81	14.52	78.65	513.92	217
BMI	kg/m^2	1,955,339	8.12	13.29	16.70	32.42	219.51	44

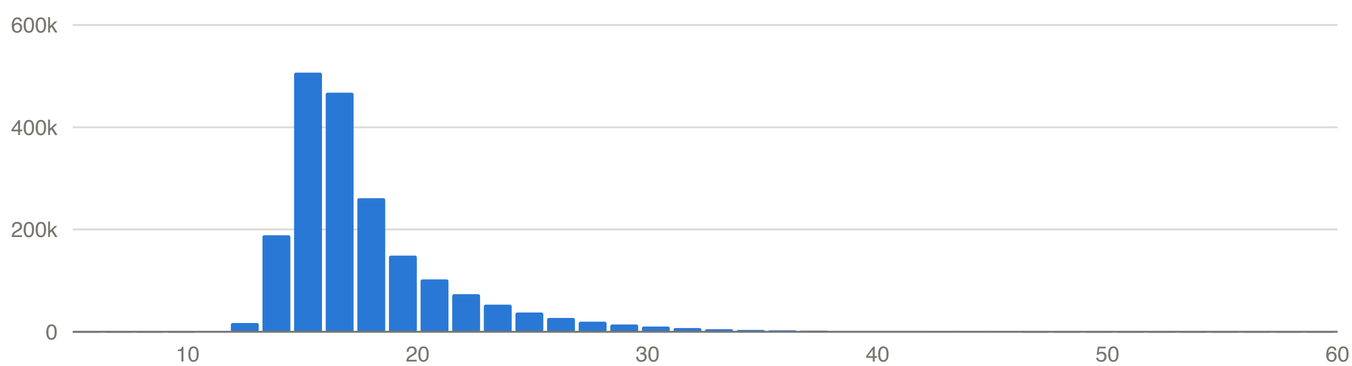
CHANNEL	UNIT	VALUES	MIN	1ST PCT	MEDIAN	99TH PCT	MAX	OUTSIDE REVIEW RANGE
head circumference	cm	1,635,690	0.00	33.00	44.00	53.00	505.46	13,742



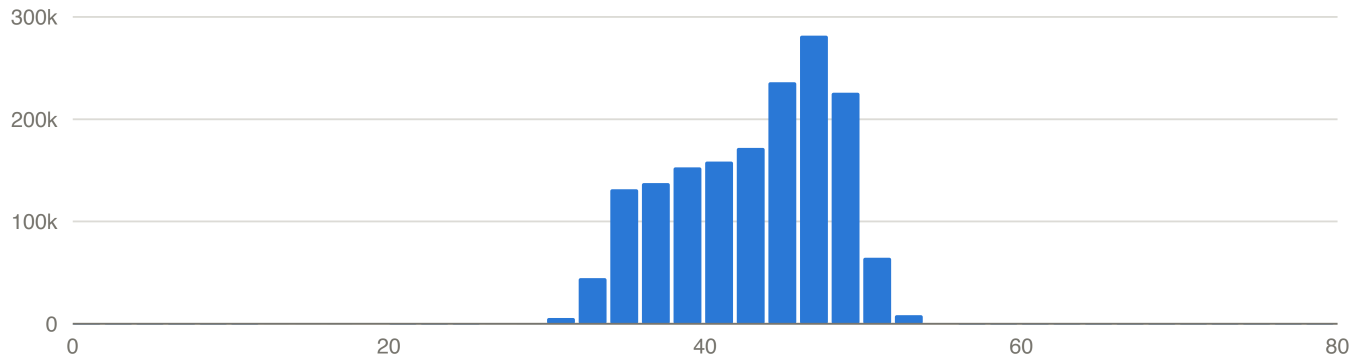
Distribution of height (cm)



Distribution of weight (kg)



Distribution of BMI (kg/m^2)



Distribution of head circumference (cm)

> **Implications for analysis.** Head circumference is the channel whose tails are worst, and 4.4 shows why. For the others the extremes are sparse but the bulk is clinically ordinary. Bound the raw imperial columns rather than the derived metric ones when screening, since a wrong unit survives an exact conversion unchanged.

4.4 Transcription-error signatures in the typed fields

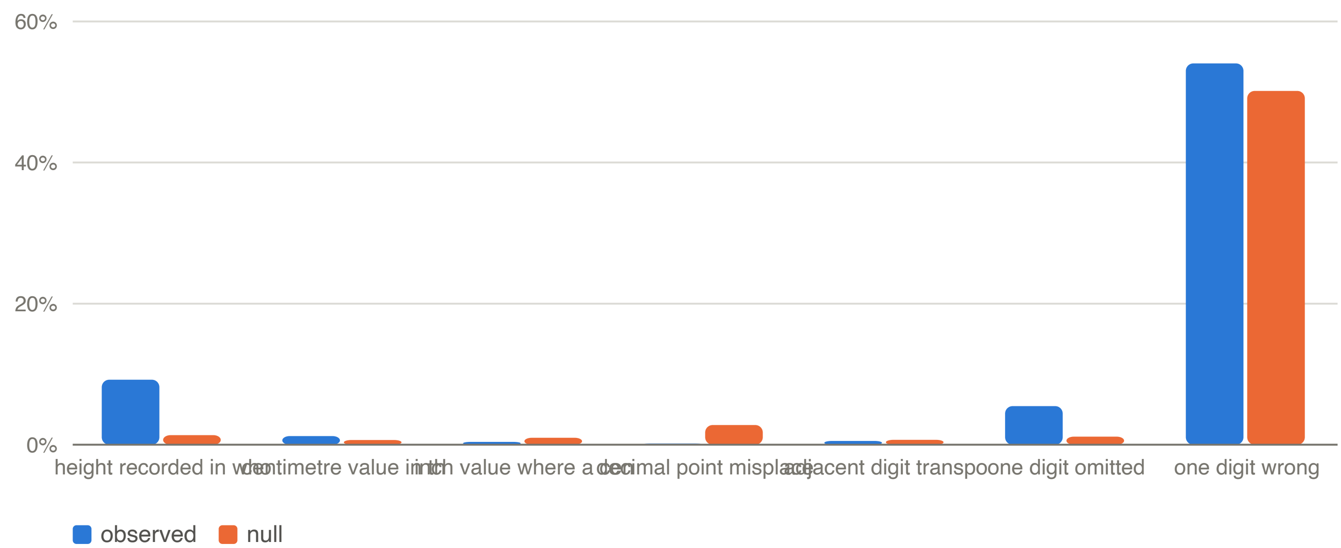
Method. Each measurement is anchored by linear interpolation between the same child's previous and next measurement. Both neighbours must themselves be plausible and span no more than four years, so a bad neighbour cannot manufacture an anomaly. A height is anomalous more than 3 inches from that anchor, a weight more than 50% from it. A mechanism *reconciles* an anomaly when applying it to the recorded value lands back at the anchor.

The null. Each anomaly's anchor is replaced by the recorded value plus a deviation drawn from another anomaly in the same year-of-age band, 20 times. That preserves the distribution of deviations exactly and destroys only the arithmetic relationship between the recorded digits and the anchor, which is the thing under test. A mechanism that reconciles anomalies no more often than it reconciles these scrambled pairs has no evidence behind it, however many hits it returns.

Height. 7,443 anomalies in the testable interior. Mechanisms are tested one at a time and are not mutually exclusive, so the rows do not sum to the total.

Height: mechanisms against the null

MECHANISM	RECONCILED	SHARE	NULL	RATIO
height recorded in whole feet	686	9.22%	1.34%	6.9x
centimetre value in the inch field	91	1.22%	0.67%	1.8x
inch value where a centimetre is expected	30	0.40%	1.00%	0.4x
decimal point misplaced	11	0.15%	2.80%	0.1x
adjacent digit transposition	40	0.54%	0.69%	0.8x
one digit omitted	407	5.47%	1.14%	4.8x
one digit wrong (calibration class)	4,024	54.06%	50.15%	1.1x



Height: observed against null, by mechanism

Adjacent digit transposition — the classic keying error, and the one most often assumed — reconciles fewer height anomalies than chance alone. The unit error is real and it is directional: a centimetre value in the inch field is enriched, while the arithmetically opposite reading sits at or below the null. That asymmetry is what a one-way data-entry confusion looks like; a spurious mechanism would be symmetric.

The dropped-digit row does not survive inspection, and it is worth showing why. Inserting a digit into a two-digit inch value always produces a three-digit one, which is never a plausible height, so the class can only fire on a value with a single-digit integer part. Among the 6,619 height anomalies whose integer part has two or more digits it reconciles 0. Its entire 5.47% is the whole-foot family reached by another route.

Two clusters are visible without any anchor at all. 1,371 visits record a `height_in` of 1 to 6 as an exact integer, median age 5.2 years — a height of 3 or 4 for a child three or four feet tall. And 143 record a `height_in` between 90 and 115, which read as inches is implausible and read as centimetres is an ordinary preschool stature at a median age of 3.1 years. The recording grid decides between the two readings: 35.0% of that cluster falls on the quarter-inch grid against 80.0% of all heights, so those values never passed through the inch-typing workflow.

Weight. 6,196 anomalies in the testable interior.

Weight: mechanisms against the null

MECHANISM	RECONCILED	SHARE	NULL	RATIO
pound value in the ounce field	39	0.63%	0.26%	2.4x
ounce value where a pound is expected	25	0.40%	0.09%	4.7x
kilogram value in the ounce field	13	0.21%	0.09%	2.5x
gram value in the ounce field	2	0.03%	0.08%	0.4x
decimal point misplaced	1,208	19.50%	1.13%	17.2x
adjacent digit transposition	77	1.24%	6.65%	0.2x
one digit omitted	927	14.96%	8.63%	1.7x
one digit wrong (calibration class)	795	12.83%	25.39%	0.5x

Transposition is again below chance, so neither channel shows evidence of digit swapping. A misplaced decimal point, which the height channel does not show at all, is the dominant weight artifact: 1,208 anomalies at 17 times the null rate, the strongest enrichment measured anywhere in this report. An ounce value has more digits than an inch value and no natural decimal point, so a factor of ten is both easy to key and hard to notice.

The calibration row is why the null is not optional. Allowing any single digit to be wrong reconciles about half of all height anomalies and reconciles almost exactly as many randomly paired values. Reported without a null it would look like the largest finding here.

How strong is the transposition negative? Only as strong as the share of transpositions the anomaly gate could have caught. Applying every adjacent digit swap to a sample of measurements in the testable interior gives that share directly: 69.7% of height swaps would displace a value past the gate, against 31.6% of weight swaps. The height negative is well powered; the weight negative rules out only large swaps, since a four-digit ounce value can absorb a swap without moving far.

What the mechanisms account for

CHANNEL	ANOMALIES	A NAMED MECHANISM FITS	ONLY THE CALIBRATION CLASS	NOTHING FITS
height	7,443	939	3,989	2,515
weight	6,196	1,987	742	3,467

Implications for analysis. Digit transposition can be dropped from the checklist for this extract at the magnitude that displaces a measurement from its own trajectory; for weight the same test is only about a third sensitive, so a small swap is not ruled out. Unit confusion and decimal placement do matter, and both are cheap to screen because both produce values implausible on their face. Bound `height_in` and `weight_oz` before any conversion, and check the recording grid rather than the value alone — the grid separates a tall adolescent from a centimetre in the wrong field where magnitude cannot. Note also that 1,371 of the whole-foot entries and 143 of the centimetre cluster already carry a null `height_cm` : the derived layer's own bound removes them as a side effect, so anyone reading the derived channels is protected and anyone reading the raw ones is not.

4.5 Repeated measurements: zero growth and apparent height loss

Children do not shrink, so a recorded decrease is recording behaviour rather than physiology. That much is easy. What matters is that the behaviour is not one thing, and the interval between measurements separates the mechanisms.

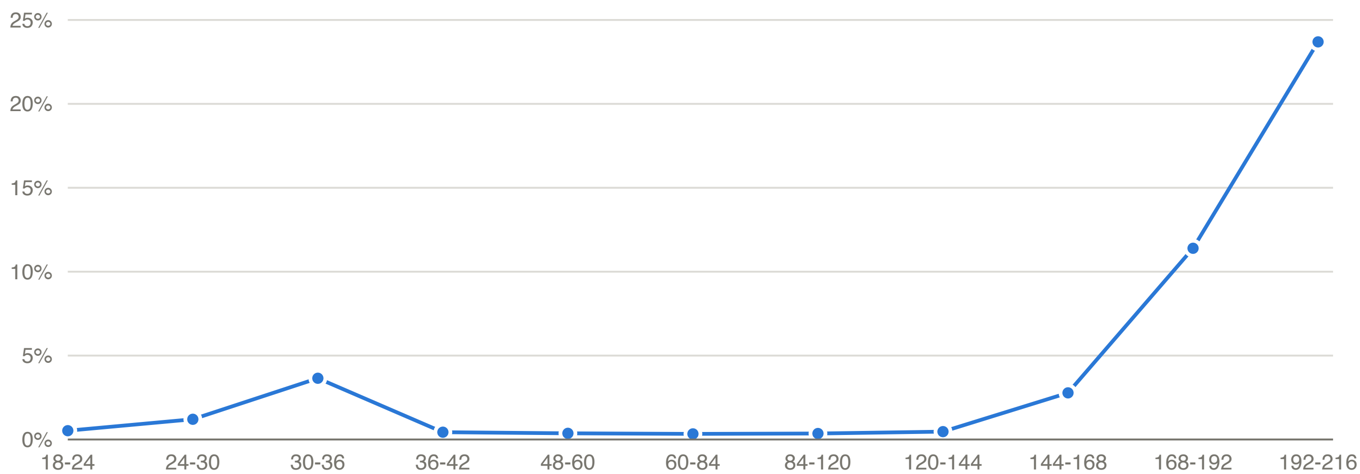
Repeat height pairs at age 2 or later, by interval

INTERVAL	PAIRS	EXACTLY ZERO CHANGE	ANY DECREASE	MEDIAN LOSS
up to 7 days	25,968	47.86%	21.31%	0.69 cm
8 to 30 days	82,652	32.00%	19.03%	0.64 cm
31 to 90 days	170,521	16.38%	10.51%	0.64 cm
91 to 180 days	205,153	5.15%	4.22%	0.81 cm

INTERVAL	PAIRS	EXACTLY ZERO CHANGE	ANY DECREASE	MEDIAN LOSS
181 to 365 days	368,248	1.57%	1.44%	0.79 cm
over 365 days	893,691	0.64%	0.66%	0.64 cm

At short intervals a child genuinely has not grown a measurable amount and the quarter-inch grid absorbs the rest. At long intervals both effects should vanish, and they do not entirely. The residue is small and its median size is about one grid step of 0.635 cm, which is the first clue that it is rounding rather than error.

Holding the interval fixed and varying age identifies the mechanisms directly.

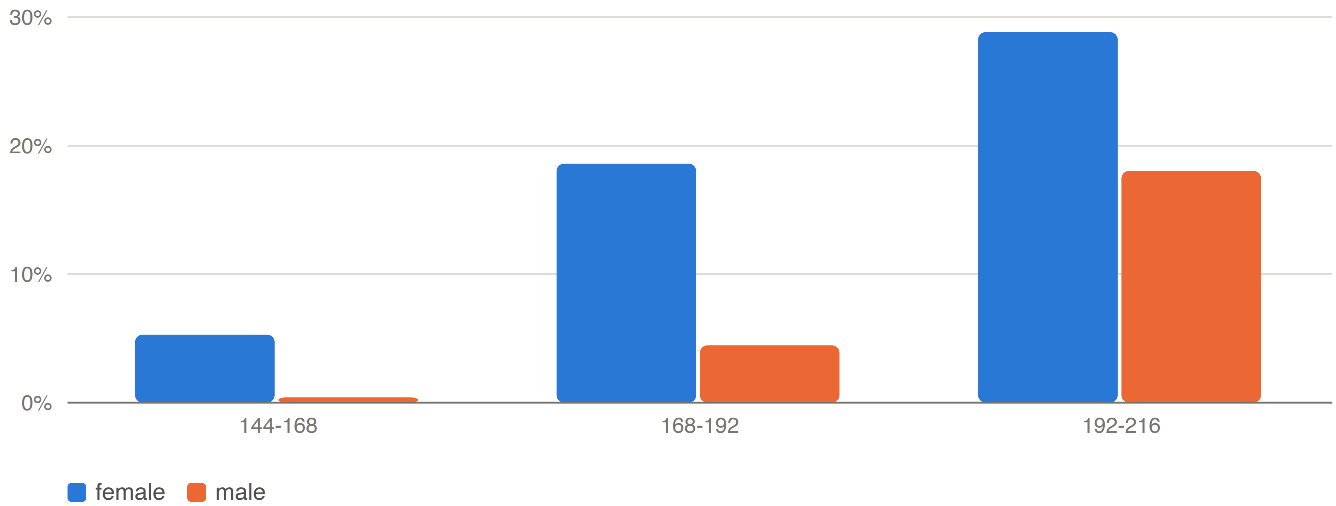


Apparent height loss by age, over 181-365 day intervals

Apparent loss by age at the earlier measurement

AGE BAND (MONTHS)	PAIRS	ANY DECREASE	MEDIAN LOSS	MEAN CHANGE
18-24	63,685	0.53%	1.27 cm	5.51 cm
24-30	52,826	1.21%	0.99 cm	5.20 cm
30-36	19,242	3.65%	1.25 cm	3.75 cm
36-42	32,650	0.44%	1.63 cm	6.19 cm
48-60	37,302	0.37%	1.91 cm	5.68 cm
60-84	64,090	0.33%	1.92 cm	5.14 cm
84-120	76,194	0.36%	1.60 cm	4.70 cm
120-144	37,388	0.47%	1.25 cm	4.87 cm
144-168	26,514	2.78%	0.64 cm	4.04 cm
168-192	14,518	11.39%	0.64 cm	1.79 cm
192-216	2,566	23.69%	0.64 cm	0.54 cm

Two separate excesses, with different signatures. The first is a narrow spike at 30 to 36 months carrying a median loss of over a centimetre — the age at which recumbent length gives way to standing height, and a standing height genuinely is shorter than a recumbent length for the same child. It is a change of measurement protocol recorded in a field that does not name the protocol.



Apparent loss in adolescence, by sex

Adolescent bands split by recorded sex

AGE BAND (MONTHS)	FEMALE PAIRS	FEMALE DECREASE	FEMALE MEAN CHANGE	MALE PAIRS	MALE DECREASE	MALE MEAN CHANGE
144-168	12,858	5.29%	2.53 cm	13,656	0.41%	5.46 cm
168-192	7,127	18.59%	0.70 cm	7,391	4.45%	2.83 cm
192-216	1,346	28.83%	0.22 cm	1,220	18.03%	0.89 cm

The second excess is the adolescent rise, and the sex split identifies it. Girls reach the high rates about two years before boys, in the same order as growth cessation, while the mean change over the same interval falls towards zero. Once annual growth drops below the recording grid, re-measuring a child who has stopped growing returns a lower value about as often as a higher one. Restricting to ages 2 to 10, where growth is unambiguously ongoing, collapses the long-interval decrease rate from 0.663% to 0.083% — 565 pairs of 681,114.

What survives both explanations divides again. Of 1,188 decreases over a centimetre across more than a year that are followed by a further measurement, 725 (61.0%) are followed by a value back at or above the earlier level, and 463 (39.0%) by one that stays below it. In the first the low value is the suspect; in the second it is corroborated and the earlier, higher measurement is the candidate error.

> **Implications for analysis.** Most apparent shrinkage here is not error and should not be filtered as an outlier: it is the recording grid acting on a flattened trajectory, plus a protocol change at two to three years. A synthetic or smoothed trajectory that lacks both will not resemble this panel. Where a decrease does need adjudication, 39% of long-interval losses persist into the next measurement, so a rule that always discards the lower value is wrong on that share.

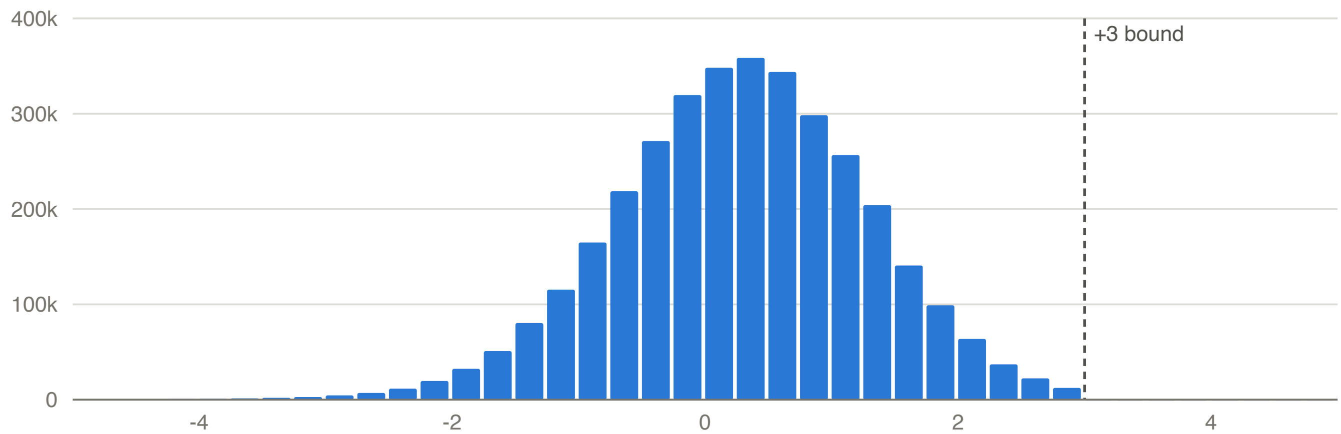
4.6 Derived z-scores and percentiles: bounds and saturation

The derived channels are not a neutral restatement of the measurements. Each carries its own support, and they do not share one.

Z-score channels

CHANNEL	VALUES	MINIMUM	MAXIMUM	BEYOND lzl
height z	3,491,616	-4.9992	3.0000	0
weight z	6,482,932	-4.9991	4.9995	0
BMI z	1,955,337	-18.7803	6.7026	400
head circ z	1,635,640	-17,485.9115	306,212.5991	16,663
weight-for-length z	2,027,317	-145.6016	7.6285	1,123
weight-for-stature z	1,371,347	-14.3445	7.4762	246

The height z-score is bounded above at exactly 3.00 while its lower tail runs past -4.99. The truncation leaves no pile-up at the boundary, so it is invisible in a summary: only 21 visits sit at or above +3. The asymmetry is what exposes it. In the lower tail 45.4% of the mass beyond lzl = 2.5 continues past 3; if the upper tail behaved the same way roughly 15,800 visits would sit above +3.



Height z-score, both tails

Percentile channels and their saturation points

CHANNEL	VALUES	EXACTLY 0	SHARE	EXACTLY 100	SHARE
height	3,491,616	2,801	0.080%	0	0.000%
weight	6,482,932	3,584	0.055%	5,221	0.081%
BMI	1,955,337	1,599	0.082%	1,590	0.081%
weight-for-length	2,027,317	6,151	0.303%	1,203	0.059%
weight-for-stature	1,371,347	1,509	0.110%	856	0.062%

Implications for analysis. The height channel cannot support any question about tall stature: its upper tail is absent, and a trajectory approaching the bound from below is distorted too. The percentile channels carry point masses at exactly 0 and 100 that are saturated rather than measured, so they are not continuous and should not be modelled as such. Because the four z channels do not share a support, a model consuming several of them together inherits the inconsistency silently. Recomputing from the raw measurement against a stated reference avoids all of this.

6. Field index

Every column, with its population, range, and the findings that govern it.

6.1 Every column in the extract

All 176 distinct columns across the 8 resources, with how much of each is populated and how many values it takes. Repeated families — the 33 encounter-diagnosis slots and the 8 race slots — appear once each, summarised on their first member.

Field index

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
patients	patient_id	VARCHAR	250,588	0.0%	250,588
patients	sex	VARCHAR	250,588	0.0%	3
patients	ethnicity	VARCHAR	245,124	2.2%	6
patients	race_1..8	VARCHAR	241,770	3.5%	11
patients_augmented	patient_id	VARCHAR	250,588	0.0%	250,588
patients_augmented	sex	VARCHAR	250,588	0.0%	3
patients_augmented	ethnicity	VARCHAR	199,143	20.5%	2
patients_augmented	race_1..8	VARCHAR	200,533	20.0%	7
patients_augmented	healthy_flag	BIGINT	250,588	0.0%	2
patients_augmented	chronic_dx_flag	BIGINT	250,588	0.0%	2
patients_augmented	growth_dx_flag	BIGINT	250,588	0.0%	2
patients_augmented	ever_stunting_flag	BIGINT	250,588	0.0%	2
patients_augmented	ever_wasting_flag	BIGINT	250,588	0.0%	2
patients_augmented	ever_underweight_flag	BIGINT	250,588	0.0%	2
patients_augmented	ever_obesity_flag	BIGINT	250,588	0.0%	2
patients_augmented	visits_count	BIGINT	250,588	0.0%	173
patients_augmented	visits_count_pre_dx	BIGINT	250,588	0.0%	170
patients_augmented	min_visit_age_days	BIGINT	250,588	0.0%	4,793
patients_augmented	max_visit_age_days	BIGINT	250,588	0.0%	6,559
patients_augmented	visits_span_days	BIGINT	250,588	0.0%	5,635
patients_augmented	dx_age_years	DOUBLE	35,890	85.7%	3,758
patients_augmented	dx_age_years_e03_9	DOUBLE	309	99.9%	283
patients_augmented	dx_age_years_e10	DOUBLE	491	99.8%	468
patients_augmented	dx_age_years_e22_0	DOUBLE	3	100.0%	3

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
patients_augmented	dx_age_years_e23_0	DOUBLE	150	99.9%	148
patients_augmented	dx_age_years_e23_6	DOUBLE	31	100.0%	31
patients_augmented	dx_age_years_e24	DOUBLE	1	100.0%	1
patients_augmented	dx_age_years_e30_0	DOUBLE	419	99.8%	365
patients_augmented	dx_age_years_e30_1	DOUBLE	3,405	98.6%	1,969
patients_augmented	dx_age_years_e34_3	DOUBLE	447	99.8%	418
patients_augmented	dx_age_years_e34_4	DOUBLE	83	100.0%	82
patients_augmented	dx_age_years_e72_11	DOUBLE	1	100.0%	1
patients_augmented	dx_age_years_k50	DOUBLE	113	100.0%	113
patients_augmented	dx_age_years_k51	DOUBLE	62	100.0%	60
patients_augmented	dx_age_years_k90_0	DOUBLE	898	99.6%	799
patients_augmented	dx_age_years_n18	DOUBLE	70	100.0%	69
patients_augmented	dx_age_years_n25_0	DOUBLE	1	100.0%	1
patients_augmented	dx_age_years_p04_3	DOUBLE	53	100.0%	41
patients_augmented	dx_age_years_p05	DOUBLE	4,069	98.4%	384
patients_augmented	dx_age_years_p07	DOUBLE	11,014	95.6%	973
patients_augmented	dx_age_years_p70	DOUBLE	3,353	98.7%	96
patients_augmented	dx_age_years_p92_6	DOUBLE	14,428	94.2%	222
patients_augmented	dx_age_years_q77	DOUBLE	15	100.0%	14
patients_augmented	dx_age_years_q78_0	DOUBLE	10	100.0%	10
patients_augmented	dx_age_years_q78_1	DOUBLE	2	100.0%	2
patients_augmented	dx_age_years_q87_1	DOUBLE	58	100.0%	57
patients_augmented	dx_age_years_q87_2	DOUBLE	32	100.0%	31
patients_augmented	dx_age_years_q87_3	DOUBLE	46	100.0%	45
patients_augmented	dx_age_years_q87_4	DOUBLE	17	100.0%	17
patients_augmented	dx_age_years_q90	DOUBLE	205	99.9%	109
patients_augmented	dx_age_years_q96	DOUBLE	36	100.0%	27
patients_augmented	dx_age_years_q98_0	DOUBLE	26	100.0%	17
patients_augmented	dx_age_years_q98_4	DOUBLE	42	100.0%	29
patients_augmented	dx_age_years_q98_5	DOUBLE	17	100.0%	13
patients_augmented	count_weight_z_score	BIGINT	250,588	0.0%	174
patients_augmented	mean_weight_z_score	DOUBLE	250,577	0.0%	45,164
patients_augmented	std_weight_z_score	DOUBLE	249,595	0.4%	15,604
patients_augmented	min_weight_z_score	DOUBLE	250,577	0.0%	47,190
patients_augmented	max_weight_z_score	DOUBLE	250,577	0.0%	47,420
patients_augmented	count_height_z_score	BIGINT	250,588	0.0%	97
patients_augmented	mean_height_z_score	DOUBLE	250,261	0.1%	41,024

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
patients_augmented	std_height_z_score	DOUBLE	248,172	1.0%	14,332
patients_augmented	min_height_z_score	DOUBLE	250,261	0.1%	42,098
patients_augmented	max_height_z_score	DOUBLE	250,261	0.1%	38,259
patients_augmented	count_bmi_z_score	BIGINT	250,588	0.0%	92
patients_augmented	mean_bmi_z_score	DOUBLE	213,053	15.0%	46,476
patients_augmented	std_bmi_z_score	DOUBLE	199,693	20.3%	15,234
patients_augmented	min_bmi_z_score	DOUBLE	213,053	15.0%	50,618
patients_augmented	max_bmi_z_score	DOUBLE	213,053	15.0%	47,254
patients_augmented	count_head_circ_z_score	BIGINT	250,588	0.0%	30
patients_augmented	mean_head_circ_z_score	DOUBLE	200,584	20.0%	46,033
patients_augmented	std_head_circ_z_score	DOUBLE	188,063	25.0%	23,188
patients_augmented	min_head_circ_z_score	DOUBLE	200,584	20.0%	31,399
patients_augmented	max_head_circ_z_score	DOUBLE	200,584	20.0%	36,528
patients_augmented	count_weight_for_length_z_score	BIGINT	250,588	0.0%	61
patients_augmented	mean_weight_for_length_z_score	DOUBLE	220,449	12.0%	42,565
patients_augmented	std_weight_for_length_z_score	DOUBLE	207,992	17.0%	18,775
patients_augmented	min_weight_for_length_z_score	DOUBLE	220,449	12.0%	42,518
patients_augmented	max_weight_for_length_z_score	DOUBLE	220,449	12.0%	38,600
patients_augmented	count_weight_for_stature_z_score	BIGINT	250,588	0.0%	63
patients_augmented	mean_weight_for_stature_z_score	DOUBLE	215,154	14.1%	45,654
patients_augmented	std_weight_for_stature_z_score	DOUBLE	203,194	18.9%	15,593
patients_augmented	min_weight_for_stature_z_score	DOUBLE	215,154	14.1%	39,544
patients_augmented	max_weight_for_stature_z_score	DOUBLE	215,154	14.1%	38,957
visits	patient_id	VARCHAR	6,494,473	0.0%	250,588
visits	visit_id	VARCHAR	6,494,473	0.0%	6,494,473
visits	age_in_days	BIGINT	6,494,473	0.0%	6,563
visits	encounter_type	VARCHAR	6,494,473	0.0%	45
visits	orig_enc_source_Epic_yn	VARCHAR	6,494,473	0.0%	2
visits	weight_oz	DOUBLE	6,488,028	0.1%	24,694
visits	height_in	DOUBLE	3,509,633	46.0%	4,836
visits	head_circ_cm	DOUBLE	1,635,690	74.8%	2,394
visits	BMI	DOUBLE	3,658,303	43.7%	8,216
visits	bmi_percentile	DOUBLE	2,961,185	54.4%	10,001
visits	enc_diag_1..33	VARCHAR	6,154,801	5.2%	6,892
visits_augmented	patient_id	VARCHAR	6,494,473	0.0%	250,588
visits_augmented	visit_id	VARCHAR	6,494,473	0.0%	6,494,473
visits_augmented	sex	VARCHAR	6,494,473	0.0%	3

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
visits_augmented	ethnicity	VARCHAR	5,401,217	16.8%	2
visits_augmented	race_1..8	VARCHAR	5,423,318	16.5%	7
visits_augmented	age_in_days	BIGINT	6,494,473	0.0%	6,563
visits_augmented	age_in_months	DOUBLE	6,494,473	0.0%	6,563
visits_augmented	age_in_years	DOUBLE	6,494,473	0.0%	6,563
visits_augmented	weight_oz	DOUBLE	6,488,028	0.1%	24,694
visits_augmented	weight_kg	DOUBLE	6,483,007	0.2%	20,665
visits_augmented	weight_outlier_flag	BIGINT	6,488,028	0.1%	2
visits_augmented	delta_weight_kg	DOUBLE	5,754,032	11.4%	5,530
visits_augmented	delta_age_in_days_weight	BIGINT	5,754,032	11.4%	3,424
visits_augmented	weight_velocity	DOUBLE	5,754,032	11.4%	6,449
visits_augmented	weight_z_score	DOUBLE	6,482,932	0.2%	81,802
visits_augmented	weight_percentile	DOUBLE	6,482,932	0.2%	10,001
visits_augmented	weight_for_length_z_score	DOUBLE	2,027,317	68.8%	70,036
visits_augmented	weight_for_length_percentile	DOUBLE	2,027,317	68.8%	10,001
visits_augmented	weight_for_stature_z_score	DOUBLE	1,371,347	78.9%	62,664
visits_augmented	weight_for_stature_percentile	DOUBLE	1,371,347	78.9%	10,001
visits_augmented	wasting_flag	BIGINT	6,494,473	0.0%	2
visits_augmented	height_in	DOUBLE	3,509,633	46.0%	4,836
visits_augmented	height_cm	DOUBLE	3,491,662	46.2%	4,568
visits_augmented	height_outlier_flag	BIGINT	3,509,633	46.0%	2
visits_augmented	delta_height_cm	DOUBLE	2,786,770	57.1%	2,683
visits_augmented	delta_age_in_days_height	BIGINT	2,786,770	57.1%	3,135
visits_augmented	height_velocity	DOUBLE	2,786,770	57.1%	6,653
visits_augmented	height_velocity_z_score	DOUBLE	1,127,289	82.6%	78,625
visits_augmented	height_velocity_z_score_ep	DOUBLE	977,101	85.0%	90,660
visits_augmented	height_velocity_z_score_ap	DOUBLE	960,949	85.2%	78,394
visits_augmented	height_velocity_z_score_lp	DOUBLE	961,074	85.2%	91,509
visits_augmented	height_velocity_percentile	DOUBLE	1,127,289	82.6%	10,001
visits_augmented	height_velocity_percentile_ep	DOUBLE	977,101	85.0%	10,001
visits_augmented	height_velocity_percentile_ap	DOUBLE	960,949	85.2%	10,001
visits_augmented	height_velocity_percentile_lp	DOUBLE	961,074	85.2%	10,001
visits_augmented	height_z_score	DOUBLE	3,491,616	46.2%	62,757
visits_augmented	height_percentile	DOUBLE	3,491,616	46.2%	9,988
visits_augmented	stunting_flag	BIGINT	6,494,473	0.0%	2
visits_augmented	head_circ_cm	DOUBLE	1,635,690	74.8%	2,394
visits_augmented	head_circ_z_score	DOUBLE	1,635,640	74.8%	66,115

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
visits_augmented	head_circ_percentile	DOUBLE	1,635,640	74.8%	10,001
visits_augmented	bmi	DOUBLE	1,955,339	69.9%	380,344
visits_augmented	bmi_z_score	DOUBLE	1,955,337	69.9%	71,202
visits_augmented	bmi_percentile	DOUBLE	1,955,337	69.9%	10,001
visits_augmented	bmi_category	VARCHAR	1,955,337	69.9%	4
visits_augmented	underweight_flag	BIGINT	6,494,473	0.0%	2
visits_augmented	obesity_flag	BIGINT	6,494,473	0.0%	2
visits_augmented	encounter_type	VARCHAR	6,494,473	0.0%	45
visits_augmented	orig_enc_source_Epic_yn	VARCHAR	6,494,473	0.0%	2
visits_augmented	enc_diag_1..33	VARCHAR	6,154,801	5.2%	6,892
labs	patient_id	VARCHAR	17,230,681	0.0%	247,271
labs	visit_id	VARCHAR	17,229,876	0.0%	2,859,084
labs	lab_order_id	VARCHAR	17,230,681	0.0%	6,578,838
labs	result_line_num	BIGINT	14,947,495	13.3%	149
labs	lab_order_date_age_in_days	BIGINT	17,230,681	0.0%	6,578
labs	lab_procedure_name	VARCHAR	17,230,681	0.0%	3,742
labs	lab_procedure_description	VARCHAR	17,230,681	0.0%	18,834
labs	lab_result_date_age_in_days	BIGINT	14,947,495	13.3%	6,613
labs	result_component_name	VARCHAR	14,947,495	13.3%	12,902
labs	result_loinc_code	VARCHAR	1,350,102	92.2%	2,194
labs	result_value	VARCHAR	14,736,420	14.5%	104,312
labs	result_flag	VARCHAR	1,679,696	90.3%	35
medications	patient_id	VARCHAR	3,823,049	0.0%	236,323
medications	visit_id	VARCHAR	3,823,049	0.0%	2,757,560
medications	med_record_id	VARCHAR	3,823,049	0.0%	3,823,049
medications	med_order_date_age_in_days	BIGINT	3,823,049	0.0%	6,605
medications	med_start_date_age_in_days	BIGINT	3,539,983	7.4%	6,644
medications	med_end_date_age_in_days	BIGINT	3,372,709	11.8%	6,939
medications	med_record_type	VARCHAR	3,823,049	0.0%	2
medications	med_simple_generic_name	VARCHAR	3,823,049	0.0%	1,073
problem_list	patient_id	VARCHAR	1,709,584	0.0%	238,823
problem_list	problem_list_id	VARCHAR	1,709,584	0.0%	1,709,584
problem_list	noted_date_age_in_days	BIGINT	1,702,300	0.4%	6,930
problem_list	resolved_date_age_in_days	BIGINT	757,907	55.7%	6,565
problem_list	pl_diag	VARCHAR	1,709,584	0.0%	4,739
referrals	patient_id	VARCHAR	349,827	0.0%	138,071
referrals	visit_id	VARCHAR	324,997	7.1%	298,615

RESOURCE	FIELD	TYPE	POPULATED	MISSING	DISTINCT VALUES
referrals	referral_id	VARCHAR	349,827	0.0%	349,827
referrals	referral_date_age_in_days	BIGINT	349,827	0.0%	6,535
referrals	requested_specialty	VARCHAR	322,375	7.8%	119
referrals	referral_number_of_visits	BIGINT	323,226	7.6%	6

7. Artifact catalogue

One row per known artifact, with its scale and whether it can be repaired.

7.1 Every artifact this report measured

One row per artifact, gathered from the findings that measured them. The class says who produced the artifact, which decides whether it can be repaired: a derivation artifact can be recomputed without touching the clinical record, a capture artifact cannot, a selection artifact is outside the extract entirely. 11 artifacts across 4 classes (capture, derivation, linkage, selection).

Artifact catalogue

ARTIFACT	CLASS	SCALE IN THIS SNAPSHOT	RECOVERABLE?	SECTION
Raw and augmented BMI disagree on infants	derivation	1,703,005 visits carry a raw BMI the augmented layer withholds; 536 differ outright	Yes — pick the layer deliberately and state which	1.3
Cohort selected on growth-measurement density and code rarity	selection	437,996 registry members reduced to 250,588; 61% of diagnosis codes, 56% of medications and 72% of lab procedures removed with their patients	No — the excluded patients are not in this extract	1.4
A patient-day can carry more than one visit	capture	5,478 patient-days holding 11,040 visits	Partly — define an explicit tie rule before ordering by age	3.1
Populated visit_id that resolves to no visit	linkage	up to 42% of populated values in a resource	No — treat visit linkage as partial by design	3.2
Age fields that violate their own ordering	capture	lab result before order, and medication start before order	No — do not treat differences between them as durations	3.3
Laboratory results are semi-structured text	capture	487,168 comparator-prefixed values; only 44.2% of rows parse as a number	Yes — parse comparators explicitly rather than casting	3.6
Anthropometrics recorded on encounters with no physical contact	capture	weight present on 99% of 22,053 telephone encounters	Partly — restrict by encounter type before counting measurement occasions	3.7
Terminal-digit heaping on the imperial recording grid	capture	80.0% of heights fall on a quarter inch	No — it is the precision the measurement actually has	4.2
Wrong-unit and decimal-place entry in the typed measurement fields	capture	1,371 whole-foot heights, 143 centimetre values in the inch field, and a weight decimal artifact enriched 17-fold	Yes — bound and repair the raw imperial columns before converting	4.4
Apparent height loss from the recording grid on a flat trajectory	capture	0.66% of pairs over a year apart, falling to 0.083% at ages 2 to 10	Not a defect — do not filter it as an outlier	4.5
Height z-score truncated above at +3 while the lower tail runs to -5	derivation	21 visits at or above +3 where roughly 15,800 would be expected	Yes — recompute from the retained raw height	4.6

8. Methods and limitations

How these figures were computed and what would invalidate them.

8.1 Methods, determinism, and limitations

Computation. Every figure was computed with DuckDB against the typed bundle of `ppoc-pediatric-ehr` 1.0.0, snapshot `2026-08-24`, sha256 `425c6f873cefc149344570561a03b33c69a6a6af7fa18bc777c0429579507116`, opened read-only. The bundle is never copied into this repository and no row-level identifier is read into any output.

Privacy. Output is aggregate only. Cells backed by fewer than 10 records are suppressed centrally rather than probe by probe, so a new probe inherits the rule without having to remember it.

Reproducibility. The generator computes the finding set once and renders every output from it, so the HTML, the PDF, the Markdown mirror, and `findings.json` cannot disagree. Prose carries templates rather than literals: a number reaches an output only by way of the finding that measured it. Outputs are rewritten only when the finding set changes, so rebuilding an unchanged snapshot leaves the committed files untouched.

Limitations. Everything here is specific to this snapshot and would need recomputing for another extract. The report describes recording and derivation behaviour, not clinical truth: a value being implausible does not establish what the child actually measured, and a value being plausible does not establish that it was measured at all. Where a mechanism is inferred rather than observed the report says so and shows the evidence.